

a What was measured

SGD phenotype annotation

Null-mutant phenotype "inviable", curated from published papers; one record per annotation, so a gene can carry several.

Includes conditionally inviable genes.

The annotation reports no medium or temperature; the loader assumes YEPD, solid, 30 °C.

GeneEssentialitySgdDataset

Costanzo et al. 2016 SGA screens

Colony size of each array strain on SGA selection medium, relative to wild type (fitness).

Four strain types: KanMX and NatMX deletions, DAmP, temperature-sensitive (TS) alleles.

Screens at 26 °C and 30 °C; deletion and DAmP values are temperature-combined (issue #410).

SmfCostanzo2016Dataset

b The record

Experiment

genotype.perturbations[0]: gene, perturbation_type, strain_id

environment: media (name, state), temperature

phenotype: is_essential = True (SGD) or fitness, fitness_std (Costanzo)

Reference

genome_reference: strain S288C

phenotype_reference: is_essential = False (SGD) or fitness = 1.0 (Costanzo)

Publication: PubMed id (SGD) or DOI (Costanzo)

Pydantic models; one LMDB entry per record in the dataset build, one Experiment node per distinct experiment in the graph.

c Query and conversion

essentiality_smf.cql: both datasets, genes in \$gene_set, solid medium

Neo4jQueryRaw: one entry per Experiment node

CompositeFitnessConverter: is_essential = True becomes fitness 0 with a fitness 1.0 reference; tagged CONVERTED_ZERO

GenotypeAggregator: every entry of one gene in one record; nothing averaged

SGD: fitness 0

Costanzo 26 °C

Costanzo 30 °C

LabelPolicy (read time): a measured fitness outranks the converted 0; 30 °C before 26 °C